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# Chapter 1: Governing Equations in Mantle Dynamics

## 1.1 Mechanics in Mantle Dynamics

Extensive studies have been conducted on fluid dynamics of a two-phase mixture. We briefly discuss the four governing equations of mechanics, drawing on a similar discussion presented by Drew and Segel (1971), McKenzie (1984) and Katz (2022).

### 1.1.1 Phase Fraction

We begin with defining the phase fraction using an indicator function, under the assumption that multiple phases cannot coexist at the same spatial location. Accordingly, the phases are considered immiscible. The indicator function is defined as

$$i(\mathbf{x}) = \begin{cases} 1 & \text{phase}_i, \\ 0 & \text{not phase}_i. \end{cases} \quad (1.1)$$

We then define the macroscopic phase fraction function of phase  $i$  within a representative volume element  $V$  as

$$\phi_i = \frac{1}{V} \int_V i(\mathbf{x}) d\mathbf{v}, \quad (1.2)$$

where  $\sum_i \phi_i = 1$ . In incompressible setting, the phase fraction may be equivalently regarded as the volume fraction occupied by each phase within a representative elementary volume, reflecting the proportion of space occupied by each phase at the macroscopic scale. Phase-averaged (or volume-averaged) quantities, also referred to as bulk quantities, are defined as follows

$$\bar{\gamma} = \sum_i \phi_i \gamma_i, \quad (1.3)$$

where  $\gamma_i$  denotes a quantity that may be scalar- or vector-valued.

### 1.1.2 Conservation Laws

We consider the conservation equation for a single-phase system. For a given volumetric quantity  $\gamma$  defined within a representative volume element  $V$ , we can write a corresponding conservation equation as:

$$\frac{d}{dt} \int_V \gamma d\mathbf{v} = - \int_{\partial V} \mathbf{f} \cdot d\boldsymbol{\sigma} + \int_V S d\mathbf{v}, \quad (1.4)$$

where  $\mathbf{f}$  represents flux travelling across the boundary of a representative volume element  $V$ . The minus sign indicates that the positive direction aligns with normal unit vector pointing outwards.  $S$ , which is source/sink term, represents the volumetric production rate of the volumetric quantity  $\gamma$ . We apply Gauss's theorem to obtain differential form,

$$\frac{\partial \gamma}{\partial t} + \nabla \cdot \mathbf{f} = S. \quad (1.5)$$

We model the partially molten mantle as a two-phase mixture consisting of liquid magma and solid mantle. The phase fraction of liquid magma is denoted by  $\phi_l$  and phase fraction of solid mantle is labelled as  $\phi_s$ , such that the relation  $\phi_l + \phi_s = 1$  is always attained.

We now rewrite the single-phase conservation law (1.5) with the bulk density defined in (1.3) as

$$\frac{\partial \bar{\rho}}{\partial t} + \nabla \cdot \bar{\rho} \mathbf{v} = 0, \quad (1.6)$$

where  $\bar{\rho} = \phi_l \rho_l + \phi_s \rho_s$ . Under the Boussinesq approximation, we neglect the compressibility of each individual phase. Furthermore, we extend the Boussinesq approximation by disregarding the density difference between the two phases, except in the body force term. Accordingly, the continuity equation for the two-phase mixture can be written as:

$$\nabla \cdot \bar{\mathbf{v}} = 0. \quad (1.7)$$

Similarly, the conservation equation for the phase-averaged momentum takes the form of

$$\frac{\partial \bar{\rho} \mathbf{v}}{\partial t} + \nabla \cdot \bar{\rho} \mathbf{v} \otimes \mathbf{v} = \bar{\rho} \mathbf{g} + \nabla \cdot \bar{\boldsymbol{\tau}} - \nabla \bar{p}, \quad (1.8)$$

where  $\boldsymbol{\tau}$  represents deviatoric stress. By neglecting inertial effects, justified by the low Reynolds number of magma and mantle, the static force balance equation reduces to:

$$\bar{\rho}\mathbf{g} + \nabla \cdot \bar{\boldsymbol{\sigma}} = 0, \quad (1.9)$$

where  $\boldsymbol{\sigma} = \boldsymbol{\tau} - p\mathbf{I}$  represents the total stress, consisting of the deviatoric component  $\boldsymbol{\tau}$  and the isotropic pressure  $p\mathbf{I}$ .

### 1.1.3 Darcy's Law

The equations governing bulk quantities (1.7) and (1.9) alone are insufficient to simultaneously determine the two distinct velocities and pressures. Therefore, an additional constitutive relationship is required to describe the behavior of the liquid phase within the porous medium. We now examine more closely the interphase forces between the solid and liquid phases. A brief overview of the derivation of the individual governing equations for each phase is provided here, while more detailed and rigorous treatments can be found in the mixture theory literature, such as Bowen (1980). We write momentum balances for each individual phase as:

$$\begin{aligned} 0 &= \phi_l \rho_l \mathbf{g} + \nabla \cdot (\phi_l \boldsymbol{\tau}_l) - \nabla(\phi_l p_l) + \mathbf{F}_l, \\ 0 &= \phi_s \rho_s \mathbf{g} + \nabla \cdot (\phi_s \boldsymbol{\tau}_s) - \nabla(\phi_s p_s) - \mathbf{F}_s, \end{aligned} \quad (1.10)$$

where  $\mathbf{F}_l = \mathbf{F}_s$  are the interphase forces between solid and liquid phases. The interphase forces between the solid and liquid phases are equal and opposite, in accordance with Newton's third law. Neglecting the deviatoric stress in the liquid phase and retaining only the isotropic pressure component, we balance the remaining stress with the viscous drag force and the equilibrium body force. Accordingly, we obtain the following relation for the relative velocity between phases, in alignment with Darcy's law

$$\phi_l(\mathbf{v}_l - \mathbf{v}_s) + \frac{k_{\phi_l}}{\mu_l}(\nabla p_l - \rho_f \mathbf{g}) = 0, \quad (1.11)$$

where  $k_{\phi_l}$  is the matrix permeability, which usually takes the quadratic form as  $k_{\phi_l} = k_0 \phi_l^2$ .

### 1.1.4 Compaction

In the context of mantle dynamics, an additional yet unique and essential relationship regulating melt migration. We now revisit the continuous equation (1.7), considering the two phases separately and incorporating the effects of melting

$$\begin{aligned}\frac{\partial \phi_l}{\partial t} + \nabla \cdot (\phi_l \mathbf{v}_l) &= \Gamma, \\ \frac{\partial(1 - \phi_l)}{\partial t} + \nabla \cdot ((1 - \phi_l) \mathbf{v}_s) &= -\Gamma,\end{aligned}\tag{1.12}$$

where  $\Gamma$  is the unknown melting rate. We now focus on the solid phase equation to get the evolving compaction relationship as

$$\frac{\partial \phi_l}{\partial t} + \mathbf{v}_s \cdot \nabla \phi_l = \Gamma + (1 - \phi_l) \nabla \cdot \mathbf{v}_s,\tag{1.13}$$

where the divergence term  $\nabla \cdot \mathbf{v}_s$  is often referred to as the compaction rate. We approximate the static compaction relation as

$$\xi_{\phi_l} \nabla \cdot \mathbf{v}_s = p_l - p_s,\tag{1.14}$$

where  $\xi_{\phi_l}$  is often referred to as bulk viscosity or compaction viscosity. The form of the compaction viscosity  $\xi_{\phi_l}$  has been the subject of extensive discussion. McKenzie (1984) popularized the concept of compaction viscosity and relating the term to the stress balance in partially molten system. Bercovici et al. (2001) extended this formulation by incorporating surface tension effects. Rabinowicz et al. (2002) further examined compaction behavior in the limit of infinitesimally small porosity. Following the approaches of Hewitt and Fowler (2008) and Katz (2008), we approximate the compaction viscosity  $\xi_\phi$  using a simplified constitutive relationship,

$$\xi_\phi = \frac{\mu_s}{\phi},\tag{1.15}$$

which highlights the fact that increased melting facilitates the compaction of the solid phase, while in the limit of no melting, compaction is entirely eliminated.

### 1.1.5 Full Static System

By combining equations (1.7), (1.9), (1.11) and (1.14), We finally arrive with the full static flow mechanical system,

$$\phi_l(\mathbf{v}_l - \mathbf{v}_s) + \frac{k_{\phi_l}}{\mu_l}(\nabla p_l - \rho_f \mathbf{g}) = 0, \quad (1.16)$$

$$\nabla \cdot (\phi_l \mathbf{v}_l + \phi_s \mathbf{v}_s) = 0, \quad (1.17)$$

$$\nabla \bar{p} - \nabla \cdot \boldsymbol{\tau}(\mathbf{v}_s) + \bar{\rho} \mathbf{g} = 0, \quad (1.18)$$

$$\mu_s \nabla \cdot \mathbf{v}_s - \phi_l(p_l - p_s) = 0, \quad (1.19)$$

where the deviatoric stress of the solid is

$$\boldsymbol{\tau}(\mathbf{v}_s) = 2\mu_s \phi_s (\mathcal{D}\mathbf{v}_s - \frac{1}{3} \nabla \cdot \mathbf{v}_s \mathbf{I}), \quad (1.20)$$

and  $\mathcal{D}\mathbf{v}_s = \frac{1}{2}(\nabla \mathbf{v}_s + \nabla \mathbf{v}_s^T)$  is the symmetric gradient. Moreover, we adjust the relative permeability  $k_{\phi_l} = k_0 \phi_l^{2+2\Theta}$  with as constant  $\Theta \in [0, 1/2]$ .  $\mu_s$  and  $\mu_l$  are dynamic viscosities for solid and liquid correspondingly. This complete static mechanical system relates the velocities and pressures of the solid and liquid phases to a given porosity distribution, thereby laying the foundation for subsequent computations.

### 1.1.6 Scaled Formulation

The Darcy equation mathematically degenerates in the regions where the porosity is zero. To handle this issue, we use the formulation developed previously by Arbogast et al. (2017). We first define pressure potentials as

$$q_l = p_l - \rho_l g z \quad \text{and} \quad q_s = p_s - \rho_l g z, \quad (1.21)$$

where  $g = |\mathbf{g}|$  is the magnitude of gravity acceleration and  $z$  is the depth, such that  $\mathbf{g} = g \nabla z$ . Note that  $\rho_l$  is used to define both potentials, so that with  $q = \bar{q} = \bar{p} - \rho_l g z$ ,

$$p_l - p_s = q_l - q_s = \frac{1}{1 - \phi_l}(q_l - q).$$



In the mechanical system, the subscript on  $\phi_l$  will be omitted. and we will simply write it as  $\phi$ . Accordingly, the solid phase fraction will be expressed as  $1 - \phi$ , emphasizing the constraint that the two phase fractions sum to unit,

$$\phi \mathbf{v}_r + \frac{k_0 \phi^{2+2\Theta}}{\mu_l} \nabla q_l = 0, \quad (1.22)$$

$$\mu_s \nabla \cdot (\phi \mathbf{v}_r) + \frac{\phi}{1 - \phi} (q_l - q) = 0, \quad (1.23)$$

$$\nabla q - \nabla \cdot \tau(\mathbf{v}_s) = -(1 - \phi) \rho_r \mathbf{g}, \quad (1.24)$$

$$\mu_s \nabla \cdot \mathbf{v}_s - \frac{\phi}{1 - \phi} (q_l - q) = 0, \quad (1.25)$$

where relative velocity  $\mathbf{v}_r = \mathbf{v}_l - \mathbf{v}_s$  and density difference  $\rho_r = \rho_s - \rho_l$ . Scaled variables are also defined,

$$\tilde{\mathbf{v}}_r = \phi^{-\Theta} \mathbf{v}_r \quad \text{and} \quad \tilde{q}_l = \phi^{1/2} q_l, \quad (1.26)$$

and the problem is reformulated as

$$\tilde{\mathbf{v}}_r + \frac{k_0 \phi^{1+\Theta}}{\mu_f} \nabla (\phi^{-1/2} \tilde{q}_l) = 0, \quad (1.27)$$

$$\mu_s \phi^{-1/2} \nabla \cdot (\phi^{1+\Theta} \tilde{\mathbf{v}}_r) + \frac{1}{1 - \phi} (\tilde{q}_l - \phi^{1/2} q) = 0, \quad (1.28)$$

$$\nabla q - \nabla \cdot \tau(\mathbf{v}_s) = -(1 - \phi) \rho_r \mathbf{g}, \quad (1.29)$$

$$\mu_s \nabla \cdot \mathbf{v}_s - \frac{\phi^{1/2}}{1 - \phi} (\tilde{q}_l - \phi^{1/2} q) = 0, \quad (1.30)$$

where the new scaled relative velocity maintain numerical stability in zero porosity region.

## 1.2 Transport of Chemical Species and Energy

With the velocities of solid and liquid phases determined, we are able to evolve our system forward in time. However, the realistic chemical compositions of mantle and magma is highly complicated and very hard to trace at the same time. Therefore, we present a simple two-species two-phase model. In this framework, we also neglect

detailed description of chemical reaction process and excluding the effect of the change of Gibbs free energies. Similar discussion can be found in extensive literatures like Spiegelman and Elliott (1993), Hewitt and Fowler (2008), Katz (2008) and Hesse et al. (2022). Both species can exist in either the solid or liquid phase. They are assumed to be immiscible in the solid state but fully miscible in the liquid phase. By again extending the Boussinesq approximation, we consider a homogenous density  $\rho$  for both solid phase and liquid phase. Figure 1.1 illustrates the coexistence of species and phases in an representative volume element. Within the representative volume

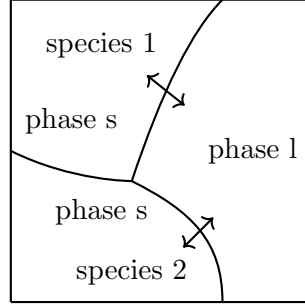


Figure 1.1: Phases and species in an representative volume element (RVE).

element  $V$ ,  $\gamma_{i,j}$  represents one quantity of the species  $i$  presenting in phase  $j$ , where  $i \in 1, 2$  and  $j \in \{s, l\}$ . The mass fraction  $c_{i,j}$  of a given species  $i$  in phase  $j$  is also regarded as concentration. We can write transport equations for a individual species  $i$  assuming a pseudo melting term  $\Gamma$  representing the transition between solid phase and liquid phase,

$$\frac{\partial}{\partial t}(\phi_l \rho_l c_{i,l}) + \nabla \cdot (\phi_l \rho_l c_{i,l} \mathbf{v}_l - \phi_l \rho_l \mathcal{D}_{i,l} \nabla c_{i,l}) = \Gamma, \quad (1.31)$$

$$\frac{\partial}{\partial t}(\phi_s \rho_s c_{i,s}) + \nabla \cdot (\phi_s \rho_s c_{i,s} \mathbf{v}_s - \phi_s \rho_s \mathcal{D}_{i,s} \nabla c_{i,s}) = -\Gamma, \quad (1.32)$$

where the diffusion term is dropped in solid phase. By summing up equation (1.31) and equation (1.32), we obtain an advection-diffusion equation expressed in terms of the phase-averaged volumetric bulk variable

$$\frac{\partial \overline{\rho c_i}}{\partial t} + \nabla \cdot (\overline{\rho c_i} \mathbf{v} - \phi_l \rho_l \mathcal{D}_{i,l} \nabla c_{i,l}) = 0, \quad (1.33)$$

where the phase averaged quantities are defined as  $\overline{\rho c_i} = \sum_{j \in \{s, l\}} \phi_j \rho_j c_{i,j}$  and  $\overline{\rho c_i \mathbf{v}} = \sum_{j \in \{s, l\}} \phi_j \rho_j c_{i,j} \mathbf{v}_j$ . The subscript  $i$  will be omitted, as only one chemical species needs to be explicitly computed in this two-species system. By extending the Boussinessq approximation such that  $\rho_l = \rho_s = \bar{\rho}$ , the species transport equation can be simplified for the following form

$$\frac{\partial \bar{c}}{\partial t} + \nabla \cdot (\bar{c} \mathbf{v} - \phi_l \mathcal{D}_l \nabla c_l) = 0. \quad (1.34)$$

We write energy balance equation in the phase-averaged internal energy form first,

$$\frac{\partial \overline{\rho u}}{\partial t} + \nabla \cdot \overline{\rho u \mathbf{v}} - \nabla \cdot \overline{K \nabla T} = \overline{\rho \mathcal{Q}} + \nabla \cdot \overline{\boldsymbol{\sigma} \cdot \mathbf{v}} + \overline{\rho \mathbf{v}} \cdot \mathbf{g}. \quad (1.35)$$

On the left-hand side, the equation contains the time-dependent rate of change of internal energy, along with advection and diffusion terms. On the right-hand side, it includes the heat source term  $\mathcal{Q}$ , mechanical work done by stress, and work against gravity. In a thermochemical equilibrium scenerio, we replace internal energy with enthalpy. We formulate the energy balance equation by neglecting external heating/cooling term  $\mathcal{Q}$  and dissipative heating following the discussion presented by Katz (2008),

$$\frac{\partial \overline{\rho h}}{\partial t} + \rho c_p \nabla \cdot (\overline{\mathbf{v}} T) = \rho L \nabla \cdot \phi_s \mathbf{v}_s + \overline{K \nabla^2 T} + \rho \alpha_p T \overline{\mathbf{v}} \cdot \mathbf{g}, \quad (1.36)$$

where the specific heat capacity  $c_p$  is used for both components in solid and liquid phase. The last term including gravitational acceleration has previously been referred to as adiabatic cooling or decompression cooling by McKenzie and Bickle (1988). We divide  $\rho c_p$  on both sides of the equation

$$\frac{\partial H}{\partial t} + \nabla \cdot (\overline{\mathbf{v}} T) = \frac{L}{c_p} \nabla \cdot \phi_s \mathbf{v}_s + \kappa \nabla^2 T + \frac{\alpha_p}{c_p} T \overline{\mathbf{v}} \cdot \mathbf{g}, \quad (1.37)$$

where  $H = \bar{h}/c_p$  is the bulk specific enthalpy and  $\kappa = \overline{K}/\rho c_p$  is the heat diffusivity.

### 1.2.1 Effective Velocity

The concept of effective velocity has long been employed in chemistry, particularly in the context of particle transport. In mantle dynamics, Spiegelman (1996)

applied effective velocity for equilibrium transport of concentration within the liquid phase. We define a corresponding effective velocity for the transport of bulk composition. To begin, we define the partition coefficient as  $D_i = c_{i,s}/c_{i,l}$ . The concentration of species  $i$  in the liquid and solid phase can be expressed in terms of the bulk composition as follows:

$$(\phi_l + \phi_s D_i) c_{i,l} = \bar{c}_i, \quad (1.38)$$

$$(\phi_l / D_i + \phi_s) c_{i,s} = \bar{c}_i. \quad (1.39)$$

By expressing  $c_{i,l}$  and  $c_{i,s}$  with bulk composition given by equations (1.38) and (1.39), we can rewrite the bulk velocity as

$$\bar{c}_i \bar{\mathbf{v}} = \phi_s \mathbf{v}_s c_{i,s} + \phi_l \mathbf{v}_l c_{i,l} = \left( \frac{\phi_l \mathbf{v}_l + D \phi_s \mathbf{v}_s}{\phi_l + \phi_s D} \right) \bar{c}_i. \quad (1.40)$$

We get the proper expression for the effective velocity, denoted as  $\mathbf{v}_{\text{eff}} = \frac{\phi_l \mathbf{v}_l + D \phi_s \mathbf{v}_s}{\phi_l + \phi_s D}$ , which is the appropriate velocity bulk composition should advecting. In the absence of melting, where  $\phi_l = 0$ , we retrieve  $\mathbf{v}_{\text{eff}} = \mathbf{v}_s$ . Conversely, in the case of complete melting, where  $\phi_l = 1$ , we obtain  $\mathbf{v}_{\text{eff}} = \mathbf{v}_l$ . Therefore, the definition of effective velocity is well defined at those two boundary scenerios. We rewrite the transport equation of species with effective velocity (1.40) and rewrite  $\bar{c}$  with  $C$  for simplicity as

$$\frac{\partial C}{\partial t} = \nabla \cdot (\mathbf{v}_{\text{eff}} C - \phi_l \mathcal{D}_l \nabla c_l) = 0. \quad (1.41)$$

### 1.3 Eutectic Phase Behavior

With the transport equations defined, we can now proceed to redistribute chemical species and energy within the system. The next step is to determine the thermodynamic state of the system, assuming thermodynamic equilibrium is attained at each time step. Recognizing that mid-ocean ridge forms in regions where temperature are generally lower, eutectic melting, where the presence of a secondary species significantly lowers the melting temperature, provides a suitable model for describing the melting behavior in these areas.

The binary eutectic system has been widely applied in studies involving chemical equilibrium, with detailed description can be found in textbook such as book by Lupis (1983). In the context of mantle dynamics, eutectic melting was being incorporated in work by Ribe (1985). Moreover, eutectic phase behavior has been applied to solidification of alloys by Wheeler et al. (1996). More recently, binary eutectic melting has also been applied to water-salt-brine system in glacier, as shown by Hesse et al. (2022).

We assume the two primary components in our partially molten system are olivine and orthopyroxene, which are highly miscible in the liquid phase, but immiscible in the solid phase. Olivine is considered the dominant component, while orthopyroxene represents the minor component. Assuming thermodynamic equilibrium is attained at each time step, we can determine phase fraction of liquid, volume fraction of each solid components and temperature on a binary eutectic phase diagram with transported enthalpy and bulk composition.

### 1.3.1 Clausius-Clapeyron Relation

In geological contexts, pressure can undergo significant variations during dynamic events such as mantle upwelling. As a result, the influence of pressure on the melting point cannot be neglected. To account for this, we approximate both the solidus and eutectic temperatures at a given pressure using the approximated Clausius-Clapeyron relationship. We assume a simple relation between as

$$\begin{aligned} T_m &= T_{m0} + \nu^{-1}(P - P_0), \\ T_e &= T_{e0} + \nu^{-1}(P - P_0), \end{aligned} \tag{1.42}$$

where  $\nu$  is the Clausius-Clapeyron constant, and  $P_0$  is the reference pressure usually set to be 0. The melting point  $T_{m0}$  and eutectic point  $T_{e0}$  refer to their respective values at standard atmospheric pressure.

### 1.3.2 Some Assumptions

In a eutectic system, the current thermodynamic state of the material can be identified using the bulk mass fraction of opx (orthopyroxene) and bulk enthalpy of the system. Let  $\mathcal{X}_i$  represents the bulk mass fraction of species  $i$  and  $\mathcal{X}_{i,e}$  denotes mass fraction of specie  $i$  in liquid at the eutectic point, which is a constant. The subscript  $i$  will be omitted in the later context, as we consider a binary system involving only two components.

We begin by introducing some assumptions regarding densities and other material properties.

- We assume no density difference between solid phase olivine and opx i.e.  $\rho_{\text{opx}} = \rho_{\text{olivine}}$ .
- We assume no density change during phase transition (extended Boussinesq approximation) i.e.  $\rho_l = \rho_s$ .
- We assume constant and uniform material thermodynamics properties like specific heat capacity and thermal expansion coefficient.

Following the assumptions, we express the bulk composition according to its definition as follows

$$\mathcal{X} = \frac{\phi_l \rho_l c_l + \phi_s \rho_s c_s}{\phi_l \rho_l + \phi_s \rho_s} = \phi_l c_l + \phi_s c_s = C = \phi_2 + \phi_l c_l. \quad (1.43)$$

We also express bulk specific enthalpy with temperature and latent heat

$$H = T + \phi_l L / c_p. \quad (1.44)$$

According to equations (1.42), the difference between melting point and eutectic point, defined as  $\Delta T = T_m - T_e = T_{m0} - T_{e0}$ , which remains constant regardless of pressure change. At any given pressure, we proceed by dividing this  $\Delta T$  to get a dimensionless formulation as

$$\frac{H}{\Delta T} = \frac{T}{\Delta T} + \frac{\phi_l L}{c_p \Delta T}. \quad (1.45)$$

Let  $\check{L} := \frac{L}{c_p \Delta T}$ ,  $\check{H} := \frac{H}{\Delta T}$  and  $\check{T} := \frac{T}{\Delta T}$ . We rewrite the enthalpy temperature relationship as

$$\check{H} = \check{T} + \phi_l \check{L}. \quad (1.46)$$

### 1.3.3 Phase Diagram

Binary eutectic melting is usually highly-nonlinear. For simplicity, we adopt a geometric parameterization of the binary eutectic diagram that can be analytically evaluated. We implement a linear eutectic phase boundary to distinguish between different phase regions within the diagram.

We begin with a brief description of eutectic phase diagram illustrated in Figures 1.2. The eutectic composition is set as  $C_e \approx 0.25$ . For illustration only, we plot bulk composition against a renormalized dimensionless temperature  $\tilde{T} - T_e/\Delta T$ , where  $\tilde{T}_m - T_e/\Delta T = 1$  and  $\tilde{T}_e - T_e/\Delta T = 0$ .

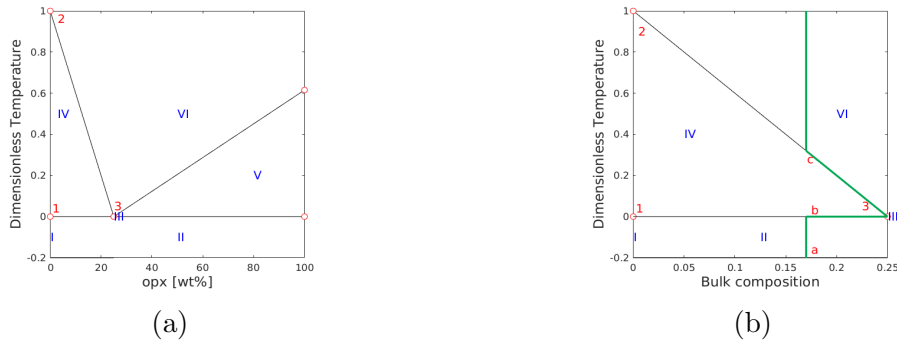


Figure 1.2: Eutectic Phase Diagram. (a) Full phase diagram of olivine-opx system. (b) Phase diagram when composition of opx smaller than eutectic composition.

We identify the following 6 regions with different phase assemblages:

- |     |                          |                                  |
|-----|--------------------------|----------------------------------|
| I   | Single-phase sub-solidus | : Olivine                        |
| II  | Two-phase sub-solidus    | : Olivine + Orthopyroxene        |
| III | Three-phase eutectic     | : Olivine + Orthopyroxene + melt |

IV Two-phase super-eutectic : Olivine + melt

V Two-phase super-eutectic : Orthopyroxene + melt

VI Single-phase super-liquidus : melt

In partial melting system, we assume the bulk composition of orthopyroxene (opx) remains below the eutectic composition. Consequently, we focus in the part of phase diagram 1.2a, where bulk composition is less than eutectic composition, which is illustrated in Figure 1.2b.

The melting in eutectic system is illustrated with the green line in Figure 1.2b. Now imagining a system containing 17% opx starts melting. Starting with all solid phase marked as point *a*, the system temperature increase linearly in response to the enthalpy rise. When the temperature reaches the eutectic temperature, marked as point *b*, both components of the system starts melting with a fixed ratio of 25% opx and 75% olivine, which is the eutectic composition. Once all opx is gone, the system leaves the eutectic points and climbs up along the line from point 3 to point *c*. After all solid exhausted, the temperature of the system will continuous to rise linearly in response to enthalpy input. It is worth noting that we can reverse the process to obtain a path of equilibrium crystallization.

#### 1.3.4 Phase Split Calculation

A detailed description of the phase-splitting calculations is provided for Regions I, II, III, and IV in this chapter. We draw a Enthalpy-Composition diagram including the effect of latent heat in Figure 1.3. Again, we plot against the renormalized enthalpy  $\check{H} - T_e/\Delta T$  for simplicity. The individual concentration and temperature are determined using equations (1.43) and (1.46). We denote the volume fraction of olivine as  $\phi_1$  and the volume fraction of orthopyroxene as  $\phi_2$ , such that  $\phi_1 + \phi_2 = \phi_s$ .



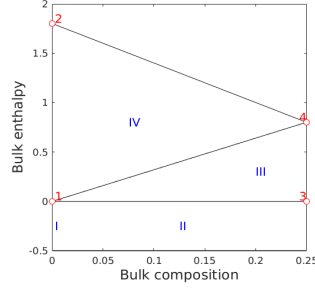


Figure 1.3: Enthalpy-Composition Diagram

Region I. Pure olivine in solid phase. The melting of pure olivine does not follow eutectic melting path. The conditions to be in region I are

$$C = 0, \quad \check{H} \leq \check{T}_m, \quad (1.47)$$

and we can determine phase fractions and temperature as

$$\phi_1 = 1, \quad \phi_2 = \phi_l = 0 \quad \text{and} \quad \check{T} = \check{H}. \quad (1.48)$$

We can determine partial derivatives of temperature with respect to bulk enthalpy and bulk composition as

$$\frac{\partial \check{T}}{\partial \check{H}} = 1 \quad \text{and} \quad \frac{\partial \check{T}}{\partial C} = 0. \quad (1.49)$$

Region II. Sub-solidus: two components coexist as solid phase. In this phase region, temperature hasn't reached eutectic melting temperature. The conditions to be in region II are

$$C \neq 0, \quad \check{H} \leq \check{T}_e, \quad (1.50)$$

and we can determine phase fractions and temperature as

$$\phi_2 = C, \quad \phi_1 = 1 - \phi_2, \quad \phi_l = 0 \quad \text{and} \quad \check{T} = \check{H}. \quad (1.51)$$

We can determine partial derivatives of temperature with respect to bulk enthalpy and bulk composition as

$$\frac{\partial \check{T}}{\partial \check{H}} = 1 \quad \text{and} \quad \frac{\partial \check{T}}{\partial C} = 0. \quad (1.52)$$

Region III. Two-phase eutectic: The system starts melting at eutectic temperature with a fixed composition ratio. Both olivine and orthopyroxene in solid state and melt coexist at the same time. Temperature remains eutectic point until opx exhausts. Concentration of opx in the melt is also fixed equal to eutectic composition. The conditions to be in region III are

$$C \leq C_e, \quad \check{T}_e < \check{H} \leq \text{line}_{1-4}, \quad (1.53)$$

where  $\text{line}_{1-4}$  is defined as  $\check{H}(C) = \check{T}_e + \check{L}C/C_e$ , and we can determine phase fractions and current temperature,

$$\phi_l = \frac{\check{H} - \check{T}_e}{\check{L}}, \quad \phi_2 = C - \phi_l C_e, \quad \phi_1 = 1 - \phi_l - \phi_2 \quad \text{and} \quad \check{T} = \check{T}_e. \quad (1.54)$$

We can determine partial derivatives of temperature with respect to bulk enthalpy and bulk composition as

$$\frac{\partial \check{T}}{\partial \check{H}} = 0 \quad \text{and} \quad \frac{\partial \check{T}}{\partial C} = 0. \quad (1.55)$$

Region IV. Super-eutectic two phase region. All solid phase opx has been exhausted. The concentration of opx in liquid  $c_l$  is now approximated with a linear function. The conditions to be in region IV are

$$C \leq C_e, \quad \text{line}_{1-4} < \check{H} \leq \text{line}_{2-4}, \quad (1.56)$$

where  $\text{line}_{2-4}$  is defined as  $\check{H}(C) = \check{H}_m - c_l/C_e$ , and we can determine phase fractions as

$$\phi_2 = 0, \quad \phi_l = \frac{C}{c_l(\check{T})} \quad \text{and} \quad \phi_1 = 1 - \phi_l. \quad (1.57)$$

The corresponding temperature can be determined by solving the following equation

$$F(\check{T}) = \check{H} - \check{T} - \frac{C}{c_l(\check{T})} \check{L} = 0. \quad (1.58)$$

Let  $c_l = C_e(\check{T}_m - \check{T})$ , we can obtain a quadratic equation about temperature as

$$\check{T}^2 - (\check{H} + \check{T}_m)\check{T} + \check{T}_m\check{H} - CL/C_e = 0, \quad (1.59)$$

which can be solved with standard quadratic root finding formula. We omit one of the roots that exceeding melting temperature,

$$\begin{aligned}\check{T} &= \frac{\check{H} + \check{T}_m - \sqrt{(\check{H} + \check{T}_m)^2 - 4(\check{T}_m\check{H} - C\check{L}/C_e)}}{2} \\ &= \frac{\check{H} + \check{T}_m - \sqrt{(\check{H} - \check{T}_m)^2 + 4C\check{L}/C_e}}{2}.\end{aligned}\tag{1.60}$$

We can determine partial derivatives of temperature with respect to bulk enthalpy and bulk composition as

$$\begin{aligned}\frac{\partial \check{T}}{\partial \check{H}} &= \frac{1}{2} \left( 1 - \frac{\check{H} - \check{T}_m}{\sqrt{(\check{H} - \check{T}_m)^2 + 4C\check{L}/C_e}} \right) \\ \frac{\partial \check{T}}{\partial C} &= - \frac{\check{L}/C_e}{\sqrt{(\check{H} - \check{T}_m)^2 + 4C\check{L}/C_e}}.\end{aligned}\tag{1.61}$$

## 1.4 Nondimensionalization

Compaction and melting are two most important aspects in mantle dynamics. We nondimensionlize the scaled formulation with characteristic length scale related to compaction length and scaled characteristic Darcy speed of the melt,

$$l_0 = \left( \frac{k_0 \mu_s}{\mu_l} \right)^{1/2} \quad \text{and} \quad u_0 = \frac{k_0 \rho_r g}{\mu_l}.\tag{1.62}$$

We define corresponding characteristic time scale as  $t_0 = l_0/u_0$ , and characeristic pressure scale as  $p_0 = \rho g l_0$ . Now we can write a full set of dimensionless equations. A set of nondimensional variables are defined as

$$\check{\mathbf{v}}_r = \frac{\mathbf{v}_r}{u_0}, \quad \check{\mathbf{v}}_s = \frac{\mathbf{v}_s}{u_0}, \quad \check{q} = \frac{q}{p_0}, \quad \check{q}_l = \frac{q_l}{p_0}.\tag{1.63}$$

| Quantity      | Value              | Units                          |
|---------------|--------------------|--------------------------------|
| $\rho$        | 3000               | $kg \cdot m^{-3}$              |
| $\rho_r$      | 600                | $kg \cdot m^{-3}$              |
| $\alpha_\rho$ | $3 \times 10^{-5}$ | $K^{-1}$                       |
| $c_p$         | 1200               | $J \cdot kg^{-1} \cdot K^{-1}$ |
| $L$           | $5 \times 10^5$    | $J \cdot kg^{-1}$              |
| $\mu_l$       | 1                  | Pascal $\cdot s$               |
| $\mu_s$       | $10^{19}$          | Pascal $\cdot s$               |
| $k_0$         | $10^{-8}$          | $m^2$                          |
| $T_{e0}$      | 1480               | $K$                            |
| $T_{m0}$      | 2000               | $K$                            |
| $\nu$         | $6.5 \times 10^6$  | Pascal $\cdot K^{-1}$          |

Table 1.1: Flow Mechanics and Thermaldynamics Parameters

The resulting dimensionless mechanics and transport equations are

$$\check{\mathbf{v}}_r + \phi^{1+\Theta} \nabla (\phi^{-1/2} \check{q}_l) = 0, \quad (1.64)$$

$$\phi^{-1/2} \nabla \cdot (\phi^{1+\Theta} \check{\mathbf{v}}_r) + \frac{1}{1-\phi} (\check{q}_l - \phi^{1/2} \check{q}) = 0, \quad (1.65)$$

$$\nabla \check{q} - \nabla \cdot \check{\tau}(\check{\mathbf{v}}_s) = -(1-\phi) \mathbf{n}, \quad (1.66)$$

$$\nabla \cdot \check{\mathbf{v}}_s - \frac{\phi^{1/2}}{1-\phi} (\check{q}_l - \phi^{1/2} \check{q}) = 0, \quad (1.67)$$

where dimensionless deviatoric stress of the solid phase is  $\check{\tau} = 2(1-\phi)(\frac{1}{2}(\nabla \check{\mathbf{v}}_s + \nabla \check{\mathbf{v}}_s^T) - 1/3 \nabla \cdot \check{\mathbf{v}}_s \mathbf{I})$ , which maintains the same form as (1.20).  $\mathbf{n}$  is the unit direction vector pointing downwards. Now we rewrite transport equations with dimensionless variables as

$$\begin{aligned} \frac{\partial C}{\partial \check{t}} + \nabla \cdot (\check{\mathbf{v}}_{\text{eff}} C - \frac{\phi \mathcal{D}_l}{u_0 l_0} \nabla c_l) &= 0, \\ \frac{\partial \check{H}}{\partial \check{t}} + \nabla \cdot (\check{\mathbf{v}} \check{T}) - \check{L} \nabla \cdot \phi_s \check{\mathbf{v}}_s - \frac{\kappa}{u_0 l_0} \nabla^2 \check{T} - \frac{\alpha_\rho l_0}{c_p} \check{T} \check{\mathbf{v}} \cdot \mathbf{g} &= 0. \end{aligned} \quad (1.68)$$

Paramters used in the flow mechanics and thermaldynamic computations are listed in Table 1.1,

## Chapter 2: Numerical Methods

We introduce a locally mass conserving numerical framework, which includes two mixed finite elements and finite volume method equipped with novel reconstruction scheme.

### 2.1 Quadrilateral Mesh

A conforming mesh of quadrilaterals will be used in both finite element and finite volume computations. Discretization on a quadrilateral mesh require fewer elements than discretization on a triangular mesh. Each quadrilateral element is assumed to be strictly convex, not degenerate into a triangle, line or point. We use a logically rectangular mesh for the ease of implementation, which can be counted with cartesian index  $\{i, j\}$ . We show a square element, which is a special case of quadrilateral mesh, and a general quadrilateral mesh in Figure 2.1. There exists a

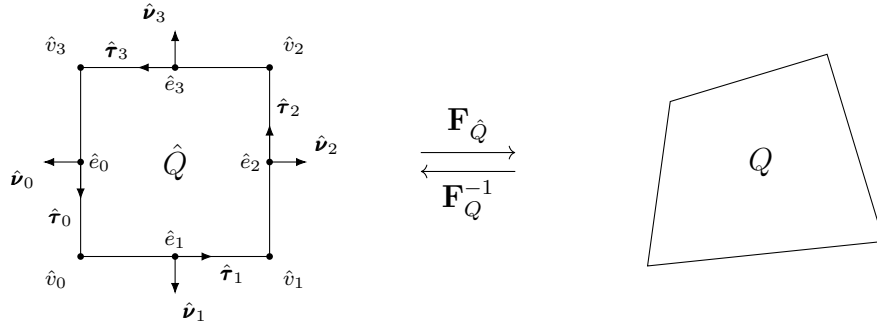


Figure 2.1: An exhibition of a square and a general quadrilateral mesh.

bijjective and smooth map  $\mathbf{F}_{\hat{Q}} : \hat{Q} \rightarrow Q$ .

## 2.2 Stokes Problem

We briefly describe solving static Stokes equation with mixed finite element scheme. We begin with strong form of the Stokes problem,

$$\begin{aligned} -\Delta \mathbf{u} + \nabla p &= \mathbf{f}, \\ \nabla \cdot \mathbf{u} &= 0. \end{aligned} \tag{2.1}$$

We derive weak form

$$\begin{aligned} (\nabla \mathbf{u}, \nabla \mathbf{v})_\Omega - (p, \nabla \cdot \mathbf{v})_\Omega &= (\mathbf{f}, \mathbf{v})_\Omega + \langle \tau^T \nabla \mathbf{u} \nu, \mathbf{v} \cdot \tau \rangle_{\Gamma_i}, \\ &+ \langle \nu^T \nabla \mathbf{u} \nu - p_0, \mathbf{v} \cdot \nu \rangle_{\Gamma_i} - \langle \nabla \mathbf{u}_0, \nabla \mathbf{v} \rangle_{\Gamma_u} \\ (\nabla \cdot \mathbf{u}, \omega) &= 0. \end{aligned} \tag{2.2}$$

Let stress  $\mathbf{s} \cdot \tau = \tau^T \nabla \mathbf{u} \nu$  and  $\mathbf{s} \cdot \nu = \nu^T \nabla \mathbf{u} \nu - p_0$ , we have traction boundary condition term defined on  $\Gamma_i$  which is  $\langle \mathbf{s}, \mathbf{v} \rangle_{\Gamma_i}$ . We employ a test space  $\{\mathbf{v} \in H^1(\Omega) : \mathbf{v} = 0 \text{ on } \Gamma_u\}$  here,

$$\begin{aligned} (\nabla \mathbf{u}_h, \nabla \mathbf{v}_h)_{\Omega_h} - (p_h, \nabla \cdot \mathbf{v}_h)_{\Omega_h} &= (\mathbf{f}, \mathbf{v}_h)_{\Omega_h} + \langle \mathbf{s}, \mathbf{v}_h \rangle_{\Gamma_{i,h}} - \langle \nabla \mathbf{u}_{0,h}, \nabla \mathbf{v}_h \rangle_{\Gamma_{u,h}}, \\ (\nabla \cdot \mathbf{u}, \omega) &= 0. \end{aligned} \tag{2.3}$$

We can separate normal and tangent part of stress on the boundary easily when physical boundary align with x-y axis. In the situation when a rectangular computational domain is used, we can mix stress and dirichlet boundary condition even further.

### 2.2.1 $H^1$ conforming discretization

## 2.3 Darcy Problem

$$\begin{aligned} \mathbf{u} + \nabla p &= \mathbf{g} \\ \nabla \cdot \mathbf{u} &= 0 \end{aligned} \tag{2.4}$$

weak form

$$\begin{aligned} (\mathbf{u}, \mathbf{v})_\Omega - (p, \nabla \cdot \mathbf{v})_\Omega &= (\mathbf{g}, \mathbf{v})_\Omega - \langle p_0, v_n \rangle_{\Gamma_i} - \langle \mathbf{u}_0, \mathbf{v} \rangle_{\Gamma_u} \\ (\nabla \cdot \mathbf{u}, \omega) &= 0 \end{aligned} \tag{2.5}$$

Test space  $\{\mathbf{v} \in H(\text{div}, \Omega) : v_n = 0 \text{ on } \Gamma_u\}$ .

### 2.3.1 $H(\text{div})$ Conforming discretization

## 2.4 Coupled Degenerate System

scaled nondimensionalized weak form let  $\check{\sigma} = \mathcal{D}\mathbf{v} - \frac{1}{3}\nabla \cdot \mathbf{v}\mathbf{I}$

$$\begin{aligned}
& (2\phi_s \check{\sigma}(\mathbf{v}_{s,h}), \nabla \Psi_s) - (\check{q}_h, \nabla \cdot \Psi_s) = -(\phi_s \check{\mathbf{f}}, \Psi_s) \\
& (\nabla \cdot \check{\mathbf{v}}_{s,h}, \omega) + \left( \frac{\hat{\phi}_f}{\phi_s} \check{q}_h, \omega \right) - \left( \frac{\hat{\phi}_f^{1/2}}{\phi_s} \check{q}_{f,h}, \omega \right) = 0 \\
& (\check{\mathbf{v}}_{r,h}, \Psi_r) - (\check{q}_{f,h}, \hat{\phi}_f^{-1/2} \nabla \cdot (\phi_f^{1+\Theta} \check{\mathbf{v}}_r)) = 0 \\
& (\hat{\phi}_f^{-1/2} \nabla \cdot (\phi_f^{1+\Theta} \check{\mathbf{v}}_r), \omega_f) + \left( \frac{1}{\phi_s} \check{q}_{f,h}, \omega_f \right) - \left( \frac{\hat{\phi}_f^{1/2}}{\phi_s} \check{q}_h, \omega_f \right) = 0
\end{aligned} \tag{2.6}$$

### 2.4.1 Degenerate Saddle Point Problem

The linear system corresponding to (insert eqcite) has the form

$$\begin{bmatrix} \mathbf{A}_s & -\mathbf{B}_s & \mathbf{0} & \mathbf{0} \\ \mathbf{B}_s^T & \mathbf{C}_s & \mathbf{0} & \mathbf{K} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_d & -\mathbf{B}_d \\ \mathbf{0} & \mathbf{K} & \mathbf{B}_d^T & \mathbf{C}_d \end{bmatrix} \begin{bmatrix} \mathbf{x}_s \\ \mathbf{y}_s \\ \mathbf{x}_d \\ \mathbf{y}_d \end{bmatrix} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{g}_s \\ \mathbf{f}_d \\ \mathbf{g}_d \end{bmatrix}, \tag{2.7}$$

wherein the solution represents the degrees of freedom of primary variables with respect to the bases of the finite element spaces. (Insert details of calculation of each blocked matrix). We form a double saddle point system by symmetrically permutate original linear system,

$$\begin{bmatrix} \mathbf{A}_s & \mathbf{0} & -\mathbf{B}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_d & \mathbf{0} & -\mathbf{B}_d \\ \mathbf{B}_s^T & \mathbf{0} & \mathbf{C}_s & \mathbf{K} \\ \mathbf{0} & \mathbf{B}_d^T & \mathbf{K} & \mathbf{C}_d \end{bmatrix} \begin{bmatrix} \mathbf{x}_s \\ \mathbf{x}_d \\ \mathbf{y}_s \\ \mathbf{y}_d \end{bmatrix} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{f}_d \\ \mathbf{g}_s \\ \mathbf{g}_d \end{bmatrix}. \tag{2.8}$$

We group these block matrix to form a concise version,

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_d \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} -\mathbf{B}_s & \mathbf{0} \\ \mathbf{0} & -\mathbf{B}_d \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} -\mathbf{C}_s & -\mathbf{K} \\ -\mathbf{K} & -\mathbf{C}_d \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{f}_d \end{bmatrix}, \quad \mathbf{G} = \begin{bmatrix} -\mathbf{g}_s \\ -\mathbf{g}_d \end{bmatrix}, \tag{2.9}$$

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^T & \mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{F} \\ \mathbf{G} \end{bmatrix}, \tag{2.10}$$

where  $\mathbf{A}$  is a positive definite matrix. However,  $\mathbf{B}$  is usually a singular matrix when no melting region presents. We will present a modified Uzawa iteration algorithm to deal with this coupled degenerate singular system. Inexact Uzawa algorithm has been introduced in the previous literature Elman and Golub (1994) and James H. Bramble (1997),

---

**Algorithm 1** Preconditioned Inexact Uzawa Iteration

---

- 1: Initialize  $\mathbf{x}_0 = \mathbf{0}$  and  $\mathbf{y}_0 = \mathbf{0}$
  - 2: Update  $\mathbf{x}_{i+1} = \mathbf{x}_i + \mathbf{A}^{-1}(\mathbf{F} - (\mathbf{A}\mathbf{x}_i + \mathbf{B}\mathbf{y}_i))$
  - 3: Obtain  $\mathbf{r} = \mathbf{B}^T \mathbf{x}_{i+1} + \mathbf{C}\mathbf{y}_i - \mathbf{G}$
  - 4: Obtain  $\tilde{\mathbf{y}} = \operatorname{argmin}_{\mathbf{y} \in V} \|(\mathbf{B}^T \mathbf{A}^{-1} \mathbf{B} - \mathbf{C})^{-1} \tilde{\mathbf{y}} - \mathbf{r}\|$
  - 5: Update  $\mathbf{y}_{i+1} = \mathbf{y}_i + \tilde{\mathbf{y}}$
- 

## 2.5 Finite Volume Scheme

## 2.6 The Multi-level Weighted Essentially Non-Oscillatory (ML-WENO) Method

We develop a accurate, versatile, robust and efficient finite volume weighted essentially non oscillatory (WENO) techniques for solving energy and composition conservation laws in Arbogast et al. (2024).

### 2.6.1 Stencil Polynomial

Given a stencil  $\mathcal{S} = \{E_0, E_2, \dots, E_k\}$  of the quasiuniform logically rectangular mesh. We can define an associated tensor product stencil polynomial  $Q_{s,t} \in \mathbb{Q}_{(s,t)} := \mathbb{P}_{s-1}(x) \otimes \mathbb{P}_{t-1}(y)$ , where  $s$  and  $t$  equal to the stencil sizes in the x- and y-directions, respectively. We scale our stencil polynomial individually as

$$Q_{s,t} = \sum_{p < s, q < t} c_{p,q} \left( \frac{x - x_s}{h_s} \right)^p \left( \frac{y - y_s}{h_s} \right)^q, \quad (2.11)$$

where  $(x_s, y_s)$  is the center point of the stencil  $\Omega_s = \cup_{E \in \mathcal{S}} E \subset \mathbb{R}^2$ , and  $h_s = \sqrt{\sum_k E_k/k}$  is the diameter of average cell size. Let  $\xi = (x - x_s)/h_s$  and  $\eta = (y - y_s)/h_s$  be the



normalized variables, we can rewrite it into normalized version:

$$\hat{Q}_{s,t} = \sum_{p < s, q < t} c_{p,q} \xi^p \eta^q. \quad (2.12)$$

In finite volume scenerio, we are dealing with cell averaged value of the scalar transport variable

$$\bar{u}_E = \frac{1}{|E|} \int_E u(\mathbf{x}) d\mathbf{x} = \frac{1}{|E|} \int_E u(x, y) dx dy. \quad (2.13)$$

We can calculate coefficients  $c_{p,q}$  by imposing a constraint that cell averaged value (2.13) is preserved through integral with stencil polynomial

$$\frac{1}{|E_k|} \int_{E_k} Q(\mathbf{x}) d\mathbf{x} = \bar{u}_{E_k}, \quad \forall E_k \in \mathcal{S}. \quad (2.14)$$

We will replace  $\bar{u}_{E_k}$  with  $\bar{u}_k$  in the following discussion for simplicity.

We can represent stencil polynomial with basis polynomials  $Q_{s,t} = \sum_k \bar{u}_k \phi_{s,t}^k$ , where basis polynomials can be calculated as

$$\frac{1}{|E_{k'}|} \int_{E_{k'}} \phi_{s,t}^k(\mathbf{x}) d\mathbf{x} = \frac{1}{|\hat{E}_{k'}|} \iint_{\hat{E}_{k'}} \hat{\phi}_{s,t}^k(\xi, \eta) d\xi d\eta = \begin{cases} 1 & k' = k \\ 0 & k' \neq k \end{cases} \quad \forall (E_{k'}, E_k) \in \mathcal{S}, \quad (2.15)$$

which would lead to a  $(s \times t) \times (s \times t)$  non-singular linear system. We can now represent stencil polynomial with normalized basis polynomials as:

$$\hat{Q}_{s,t}(\xi, \eta) = \sum_k \bar{u}_k \hat{\phi}_{s,t}^k(\xi, \eta). \quad (2.16)$$

We define a linear projection operator  $\pi_S$  and let  $\pi_S u_S = Q$  be a linear projection onto polynomial space  $W_p$  such that  $Q_{s,t} \subset W_p$ . We estimate interpolation error with the Bramble-Hilbert Lemma Dupont and Scott (1980) and Bramble and Hilbert (1970) in  $H_r(\Omega)$  space, where  $r = s$  in x direction and  $r = t$  in y-direction.

**Lemma 2.1.** *Let  $\Omega_S$  be the domain covered by stencil  $\mathcal{S}$  in  $\mathbb{R}^2$  and let  $u \in H^r(\Omega_S)$ . We have:*

$$\|u - \pi_S u\|^2 \leq h^{2(r-1)} |u|_{H^r(\Omega_S)}^2. \quad (2.17)$$

*Proof.*

$$\|u - \pi_S u\| = \quad (2.18)$$

□

### 2.6.2 Stencil polynomial smoothness indicators

We define a smoothness indicator  $\sigma$  as a measurement of smoothness of  $u(\mathbf{x})$  on the stencil  $S$ . The classic method from Jiang and Shu (1996) is defined by Sobolev seminorm as:

$$\sigma_P = \sum_{1 \leq |\alpha| \leq m} \int \int_{E_0} |E_0|^{|\alpha|-1} (\mathcal{D}^\alpha Q(x, y))^2 dx dy, \quad (2.19)$$

where  $\alpha = (\alpha_1, \alpha_2)$  is a multi-index,  $|\alpha| = \alpha_1 + \alpha_2$ , and  $\mathcal{D}$  is the derivative operator  $\mathcal{D}^\alpha = \partial^{|\alpha|} / \partial x^{\alpha_1} \partial y^{\alpha_2}$ . The Jiang-Shu smoothness indicator can be reformed with basis polynomials:

$$\begin{aligned} \sigma_P &= \sum_{1 \leq |\alpha| \leq m} |E_0|^{|\alpha|-1} \iint_{E_0} (\mathcal{D}^\alpha Q(x, y))^2 dx dy \\ &= \sum_{1 \leq |\alpha| \leq m} \iint_{\hat{E}_0} (\mathcal{D}^\alpha \hat{Q}(\xi, \eta))^2 d\xi d\eta \\ &= \sum_{1 \leq |\alpha| \leq m} \iint_{\hat{E}_0} \left( \sum_{ij} \bar{u}_{ij} \mathcal{D}^\alpha \hat{\phi}_{i,j}(\xi, \eta) \right)^2 d\xi d\eta \\ &= \sum_{ij} \sum_{i'j'} \bar{u}_{ij} \bar{u}_{i'j'} \sum_{1 \leq |\alpha| \leq m} \iint_{\hat{E}_{ij}} \mathcal{D}^\alpha \hat{\phi}_{ij}(\xi, \eta) \mathcal{D}^\alpha \hat{\phi}_{i'j'}(\xi, \eta) d\xi d\eta \\ &= \sum_{ij} \sum_{i'j'} \bar{u}_{ij} \bar{u}_{i'j'} \sigma_{ij}^{i'j'}, \end{aligned} \quad (2.20)$$

where  $\sigma_{ij}^{i'j'}$  can be precomputed.

We proposed an alternative way of defining smoothness indicator in Huang et al. (2023).

$$\sigma_{\text{simp}} = \frac{1}{N_S - 1} \sum_k (\bar{u}_k - \bar{u}_0)^2, \quad (2.21)$$

where  $\bar{u}_0$  is the cell averaged solution in target cell. We define this simple smoothness indicator in order to reduce the computational cost and to deal with distorted stencil

on unstructured mesh. However, this simple smoothness indicator is not as robust as the classic Jiang-Shu smoothness indicator. Hence, we stick to the classic definition in the following discussion.

We observe that the smoothness indicator exhibit following convergence property that there exists some value  $C > 0$  such that when  $h \rightarrow 0^+$ ,

$$\sigma = \begin{cases} Ch^2 + \mathcal{O}(h^3) & \text{Smooth,} \\ \Theta(h) & \text{Not Smooth,} \end{cases} \quad (2.22)$$

where stencils are defined on quasi uniform mesh. This observation comes directly from the scaling property of Sobolev seminorm.

### 2.6.3 ML-WENO weighting

Given a target cell  $E_{i,j}$ , we can create a nonempty set of stencils  $\mathcal{S}_l \ni E_{i,j}, l = 0, 1, \dots, L$ . The reconstruction of  $u$  on target cell  $E_{i,j}$  is a convex combination of stencil polynomials (2.16)

$$R(\mathbf{x}) = \sum_l \tilde{\omega}_l Q^l(\mathbf{x}), \quad \mathbf{x} \in E_{i,j}, \quad (2.23)$$

where  $\sum_l \tilde{\omega}_l = 1$  is normalized nonlinear weights associated with stencils accordingly. We begin by arbitrarily choosing a set of linear weights  $\{\omega_l\}$  and scale with stencil

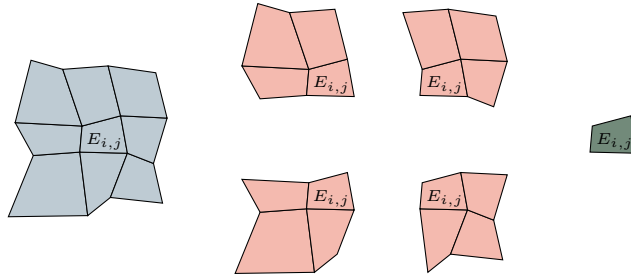


Figure 2.2: An example of (3,2,1) multi-level stencils combination on logically rectangular quadrilateral mesh.

smoothness indicator as

$$\hat{\omega}_l = \frac{\omega_l}{(\sigma_l + \epsilon h_0^2)^{ar_l + n_l}}, \quad (2.24)$$

where  $r_l = \max\{s, t\}$ ,  $\epsilon = 1e - 4$  and  $a > 0.5$  are picked as constants. The biasing factor  $n_l = n(r_l)$  is used to biasing away further from constant level, whose smoothness indicator is always zero, when the solution is smooth. We pick

$$n(r_l) = n_l = \begin{cases} 1, & \text{if } r_l = 1, \\ 3, & \text{if } r_l = 2, \\ 4, & \text{if } r_l \geq 3. \end{cases} \quad (2.25)$$

We then renormalize scaled weights

$$\tilde{\omega}_l = \frac{\hat{\omega}_l}{\sum_{l'} \hat{\omega}_{l'}}. \quad (2.26)$$

#### **2.6.4 Boundary Treatment**

### **2.7 Time Stepping**

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